

Necessity Entrepreneurs,^{*}

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Abstract

We develop a structural framework in which necessity entrepreneurship emerges endogenously from occupational choice under incomplete insurance. Households choose between employment, unemployment, and entrepreneurship in an environment with unemployment risk, search-and-matching frictions, and heterogeneous human capital and managerial ability. Production follows a span-of-control technology, so occupational sorting determines both who becomes an entrepreneur and the scale at which firms operate. Within this environment, we provide a theory-based definition of a necessity entrepreneur: an agent who chooses entrepreneurship when non-employed but would choose paid work if an employment relationship were available. This definition is grounded in primitives and incentives rather than survey labels or ex post classifications. The model characterizes how labor-market risk and insurance provision shift the boundary between opportunity and necessity entrepreneurship, with implications for average firm size and the composition of entry. Quantitatively, the framework is designed to discipline policy experiments that distinguish changes in the level of entrepreneurship from changes in its type.

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1 Introduction

Entrepreneurship is often treated as a driver of innovation, productivity, and growth. However, not all entrepreneurship has the same economic role. In many settings, a substantial share of entry is necessity-driven rather than opportunity-driven. This distinction matters because necessity entry can reflect weak labor-market outside options rather than comparative advantage in managing production.

Empirical work typically separates necessity and opportunity entrepreneurship using survey responses or prior labor-market status. Both approaches are useful, but each has limits. Self-reported motives are subjective, while classifications based only on previous employment status can misclassify agents whose timing is strategic or constrained by insurance and search frictions. A worker who expects separation may enter early and look “opportunity” in the data; an unemployed household can enter temporarily while searching and look “necessity” even when underlying talent is high.

This paper provides a model-based definition that maps directly into counterfactual behavior. We define opportunity entrepreneurs as households that choose entrepreneurship regardless of whether they enter the period employed or non-employed. We define necessity entrepreneurs as households that choose entrepreneurship only when they lack an employment relationship. The model combines heterogeneous managerial talent, human-capital heterogeneity, search frictions, and social insurance, allowing us to quantify how labor-market risk and policy jointly shape the composition of entrepreneurship.

Our contribution is threefold. First, we provide a structural framework in which necessity entrepreneurship emerges endogenously from occupational choice under incomplete insurance. Second, we provide a quantitative decomposition of how unemployment risk and unemployment insurance affect entry margins and firm scale across education groups. Third, we reinterpret necessity entry as a misallocation channel: low outside options can push households into entrepreneurship even when their productivity is better used in paid work.

The rest of the paper proceeds as follows. Section 2 discusses related literature. Section 3 presents the economic environment. Section 4 characterizes household behavior and equilibrium. Section 5 discusses calibration. Section 6 presents baseline quantitative results. Section 7 studies policy counterfactuals, and Section 8 concludes.

2 Literature Review

This paper connects three strands in a single quantitative framework.

First, we build on work that separates opportunity and necessity entrepreneurship in the data. Fairlie and Fossen (2019) propose an operational distinction based on pre-entry labor market status and show that opportunity and necessity entrepreneurship have different cyclical patterns. Sedlacek and Sterk (2017) show that startup cohorts differ in growth potential across

the business cycle. This maps directly to our decomposition of entrepreneurial entry by prior employment state.

Second, we connect to the entrepreneurship-and-frictions literature. Buera (2009) shows in a dynamic model that borrowing constraints can materially change both entry and firm scale, with large welfare effects through undercapitalized businesses. Hurst and Pugsley (2011) documents that many small businesses are not primarily innovation-driven and that nonpecuniary motives are quantitatively important. This evidence supports modeling entrepreneurship as a broad occupational margin rather than only a high-innovation margin.

Third, our labor-market block follows the search-and-matching tradition. Petrongolo and Pissarides (2001) provides the aggregate matching-function synthesis that motivates our use of separation and job-finding transitions to discipline occupational flows.

Relative to these strands, our contribution is to embed an explicit opportunity-versus-necessity distinction in a heterogeneous-agent model with search frictions, incomplete insurance, and policy counterfactuals. This structure lets us quantify how unemployment risk and social insurance jointly shift entrepreneurial entry, firm scale, and composition.

3 The Model: Economic Environment

Consider a Lucas (1978) span of control model, where individuals differ in the level of human capital, h . As in Gomez and Kuhen (2017), individuals are educated to either the primary (P), secondary (S) or tertiary level (T). The level of education means that a tertiary educated individuals have a higher draw of human capital, on average, than secondary educated individuals, who in turn have a higher draw on average than primary educated individuals. This human capital draw is positively correlated with managerial ability x as well as labor productivity z .

At the start of the period agents enter as either unemployed ($i_w = 0$), employed ($i_w = 1$), or an entrepreneur ($i_w = 2$). Individuals face a choice of whether to become an entrepreneur, to continue working or to search for work. However, the value of employment is different depending on whether an agent enters the period as employed ($i_w = 1$) or not. Although there is a risk that an employee may become separated from a firm with probability ρ , this is less risky than searching for work which has a $p(\theta)$ probability of obtaining employment. Consequently, the expected value of employment is higher for those already in employment than those who are not. Additionally, the probability of separation is lower for individuals with higher levels of human capital and they are able to command a higher wage due to higher levels of productivity. So the value of employment, holding other things equal, is higher for those with higher human capital as well.

The innovation of this model is that the entrepreneurial decision results in two types of entrepreneurs emerge, opportunity-based entrepreneurs and necessity-based entrepreneurs. We define opportunity-based entrepreneurs are those whose value of entrepreneurship is higher, even

if they had employment. Necessity-based entrepreneurs are those who would not have become entrepreneurs, if they had employment.

Timing: The timing of events can be summarized below. Occupational choice happens at the start of time t . Those choosing to become entrepreneurs run projects and hire workers on on rolling contracts (i.e. contracts, will be reassessed each period). Those who were previously employed within a project can decide to carry on working but carry a risk of separation with probability ρ . If separation occurs these workers will claim unemployed benefits and be unable to look for work until the following period. After separation occurs, vacancies are posted and who were unemployed in the previous period are matched with firms. Those that are successful and match with with probability $p(\theta)$ go on to earn a wage. Whereas those that are unsuccessful will claim unemployment benefits and be unable to search again until the following period.

Unlike the search and matching literature there is no wage bargaining between workers and the firm at the end of matching. Instead the wage pins down the occupational choice decision which determines those wanting to become entrepreneurs and those who will search for work. Crucially, the worker only finds out whether they enter production and earn a wage after their occupational choice decision.

Figure 1: Timing for Matched Workers

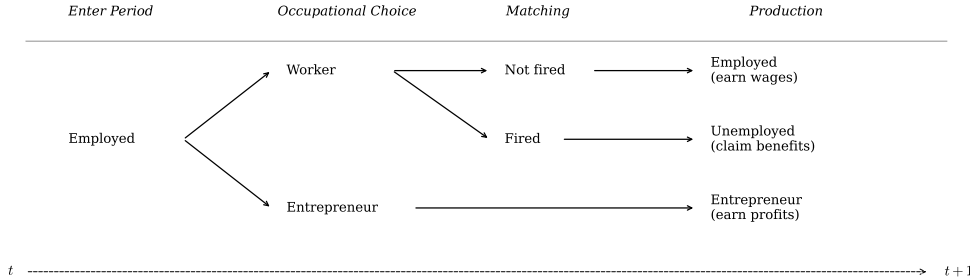
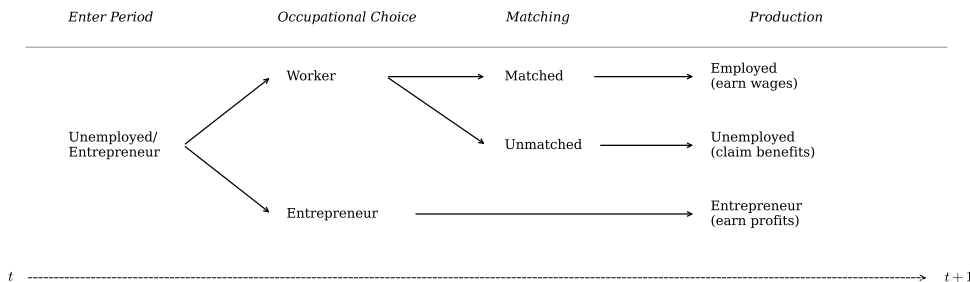


Figure 2: Timing for Unmatched Workers



3.1 Preferences, endowments and technology

Preferences: Consumption by an agent in period t is c_t , with utility given by $U(c_t)$.

Endowments: Agents are endowed with an initial capital asset, a_0 , which can be used as an input in production. Each individual is endowed with human capital shock h^i . Human capital h is positively correlated with managerial talent x , labour productivity z and separation rate ρ .

Production: Firms use labor n and capital k to produce a single consumption good, y . Capital depreciates at a constant rate of δ . Managers can operate only one project. The functional form of the production function is:

$$y = xk^\alpha(zn)^\gamma \text{ where } \alpha, \gamma > 0. \quad (1)$$

Managerial talent x is determined by the Pareto distribution $F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\vartheta}$ where ϑ is the shape parameter, ς the location parameter and ξ is the scale parameter.

Factor remuneration: Firms rent capital at the common market rate r .

Government: The government runs a balanced budget each period and provides unemployment insurance which is financed through lump sum taxation τ , which is shared by the employee $(1 - \psi)\tau$ and employer $\psi\tau$. This program guarantees each household an unemployment benefit of b .

3.2 The Firm's problem

The firm's problem is:

$$\max_{n,k} Xk^\alpha n^\gamma - (1 + \tau\psi)wzn - rk - \kappa \frac{\rho zn}{q(\theta_t)} \quad (2)$$

employees must pay a share of unemployment insurance τ to all workers who produce output. Firms have to pay search costs κ for all vacancies posted. The number of vacancies posted is designed to replace any lost workers, ρzn . This, however, is affected by the firm-hiring rate $q(\theta_t) = \bar{m} \left(\frac{v_t}{u_t}\right)^{-\mu}$, where aggregate vacancies v_t is simply the aggregate level of all those who have separate from firms $\sum \rho zn$ within the period and u_t is the total unemployed looking for work within that period (see appendix).

When the market is tight and the hiring rate is low, firms have to search more and thus face higher search costs in order to replace lost workers. This is because $0 < q(\theta_t) < 1$, so a lower $q(\theta_t)$ increases $\frac{\rho zn}{q(\theta_t)}$. If the firm lose 10 workers, they want to hire 10 workers to replace them. But if the hiring rate, $q(\theta_t)$, is 50% then they will face double the amount of search costs than

if $q(\theta_t)$ were 100%. The upshot of this is that when markets are tight, the size of the desired firm size n decreases ex ante, as the cost of replacing workers is high and so entrepreneurs would prefer to run smaller firms.

To avoid issues of directed search, we assume that employees are randomly matched with firms (as in Chivers et al., 2015) and each firm employees a distribution of workers across the human capital spectrum such that each firm employs the market average.

4 Optimal behavior and equilibrium

In any time period t agents are distinguished by (h, a, x, i, z) . The timing of the economy is given as follows.

1. Households enter each new period with assets a and an employment status i_w
2. Idiosyncratic shocks h is drawn which determines x (managerial ability) and z (labor productivity), ρ separation rate conditional on education level, P, S or T .
3. Households make an occupation decision: entrepreneur ($\mathcal{I}_e = 1$) or worker ($\mathcal{I}_e = 0$).
4. Capital and labor markets clear. A fraction ρ of employees separate from firms and $p(\theta)$ are hired to replace them
5. Households choose: consumption (c), borrowing/saving (a')

4.1 Firm manager

Firms are distinguished by their productivity realization x . Agents with sufficient ability to become managers choose the level of capital and the number of employees to maximize profit subject to a technological constraint. In order to simplify the exposition, first consider the problem of a manager with talent x^i for a given level of capital k (i.e., labor input choice only):

$$\max_{zn} xk^\alpha (zn)^\gamma - zw(1 + \psi\tau)n - rk - \kappa \frac{\rho zn}{q(\theta)} \quad (3)$$

The first order conditions are:

$$\gamma xk^\alpha zn^{\gamma-1} - w(1 + \psi\tau) - rk - \frac{\rho\kappa}{q(\theta)} \quad (4)$$

$$n^*(k, x, \tilde{w}) = \left[\frac{\gamma xk}{(w(1 + \psi\tau) + \frac{\rho\kappa}{q(\theta)})} \right]^{\frac{1}{1-\gamma}} \quad (5)$$

Substituting (5) into (3) yields the manager's profit function for a given level of capital:

$$y(k, x, \tilde{w}) = xk \left[\frac{\gamma x k^\alpha}{(w(1 + \psi\tau) + \frac{\rho\kappa}{q(\theta)})} \right]^{\frac{\gamma}{1-\gamma}} \quad (6)$$

4.1.1 Capital

Initially we assume capital markets function perfectly.

4.2 Entrepreneurs

Entrepreneurs maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + \nu - \tau_\pi \quad (7)$$

$$a' \geq -\bar{a}$$

4.3 Employed

The workforce (previous periods employed who choose to reapply for work) maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + (1 - \rho)(wz(1 - (1 - \psi)\tau_y)) + \rho b \quad (8)$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

4.4 Unemployed

The unemployed maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + p(\theta) (wz(1 - (1 - \psi)\tau_y)) + (1 - p(\theta))b \quad (9)$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

4.5 Government

The government runs a balanced budget with a lump-sum taxes $T(y) = (y - \lambda_p y^{1-\tau_p})$

4.6 The household's problem

Let i_w denote the household's labor market status in the previous period, with $i_w = 2$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 0$ being unemployed. Let $\mathcal{I}_e$ indicate occupational choice, where $\mathcal{I}_e = 1$ if the household is an entrepreneur and $\mathcal{I}_e = 0$ if the household is a worker.

Let denote $\mathbb{E}_\rho$ the expected value considering the possibility of being separated from the job as employed; $\mathbb{E}_{p(\theta)}$ the expected value considering the possibility of finding new job if previously unmatched with a firm (unemployed former entrepreneurs). V_w is the contemporary utility as a worker, which varies with the employment status. $\iota_{i_w=1}$ is an indicator function. We can write the household's problem recursively as follows. More specifically,

$$\begin{aligned} \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) &= p(\theta) (V_n + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 1)) \\ &\quad + (1 - p(\theta)) (V_u + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0)) \end{aligned}$$

$$\begin{aligned} \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) &= \rho (V_u + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0)) \\ &\quad + (1 - \rho) (V_n + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 1)) \end{aligned}$$

Notice that for agents who recently separated from the firm cannot rematch with a new firm in the current period.

$$\mathbf{V}(a, h, i_w) = \max_{\{a', c, p(\theta), \mathcal{I}_e, \mathcal{I}_w\}} \left\{ \begin{array}{l} \iota_{i_w=0} [(1 - \mathcal{I}_e) \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) + \mathcal{I}_e (V_e + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0))] + \\ \iota_{i_w=1} [(1 - \mathcal{I}_e) \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) + \mathcal{I}_e (V_e + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0))] + \\ \iota_{i_w=2} [(1 - \mathcal{I}_e) \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) + \mathcal{I}_e (V_e + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0))] \end{array} \right\}$$

subject to

$$c + a' \leq (1 - \tilde{r} - \delta)a + inc \quad (10)$$

$$inc = \begin{cases} ra + \tilde{w}_n & \text{if } \mathcal{I}_e = 0, i_w = 1 \\ ra + \tilde{w}_u & \text{if } \mathcal{I}_e = 0, i_w = 0 \\ ra + \nu(k, x; \tilde{r}, w, \tau) & \text{if } \mathcal{I}_e = 1 \end{cases}$$

where inc is the earnings of the worker or entrepreneur, $\tilde{w}_n = (1 - \rho) \{zw(1 - (1 - \psi)\tau)\} + \rho b$ and $\tilde{w}_u = p(\theta) \{zw(1 - (1 - \psi)\tau)\} + \{1 - p(\theta)\}b$.

4.7 Equilibrium

Let $\hat{h}$ be the critical value of human capital that determines occupational choice which is affected by both employment status and education level. If $h \geq \hat{h}$ then individuals choose entrepreneurship over becoming a worker. The critical value of human capital is always higher for those who have employment than those who are unmatched with a firm $\hat{h}_{i_w \neq 1} < \hat{h}_{i_w = 1}$. We define necessity entrepreneurship as those who would not have become an entrepreneur *if* they had an employment contract:

$$\hat{h}_{i_w \neq 1} < h < \hat{h}_{i_w = 1}.$$

We define opportunity-driven entrepreneurship as those who would become an entrepreneur *regardless* of circumstance:

$$h \geq \hat{h}_{i_w = 1} \geq \hat{h}_{i_w \neq 1}.$$

As there are three education levels, there are a total of six critical values of human capital: two for primary educated matched or unmatched, $\hat{h}_{i_w = 1}^P$ and $\hat{h}_{i_w \neq 1}^P$ two for secondary educated, $\hat{h}_{i_w = 1}^S$ and $\hat{h}_{i_w \neq 1}^S$, and two for tertiary educated $\hat{h}_{i_w = 1}^T$ and $\hat{h}_{i_w \neq 1}^T$.

5 Calibration

Table 1: Parameter values, baseline economy

Parameters	Values	Description	Comments/observations/target
β	0.94	Discount factor	
σ	2.0	Risk aversion	$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$
δ	0.06	Capital depreciation	
α	0.3207	Capital share	
γ	0.4693	Labour Share	Frac. of entrepreneurs
h	{0.109, 0.581, 0.310}	Share of population by h	Barro and Lee (2013)
ϑ_e	{0.967, 1.007, 1.156}	Shape parameter by h	Mean firm size, Gomez and Kuhen (2017)
ς	2.02655	Location parameter	$F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\alpha_i}$
ξ	0.1115	Scale parameter	Gomez and Kuhen (2017)
z	{1.0, 1.31, 1.61}	Wage premium by h	Goldin and Katz (2009)
ρ	{0.0383, 0.0288, 0.0136}	Job separation rate by h	Cairo and Cajner (2016)
$\bar{m}$	0.8	Matching efficiency	Lubik, 2013 $p(\theta) = \bar{m} \left(\frac{v}{u}\right)^{1-\mu}$
μ	[0.5, 0.7]	Matching elasticity	Petrongolo and Pissarides (2001)
κ	[0.03, 0.045] $\cdot w$	Search costs	(Hagedorn and Manovskii, 2008)
τ_π	0	Corporation tax	Set as zero for now
τ_y	0.151	Income Tax	$T(y) = (y - \lambda_p y^{1-\tau_p})$
λ_p		Income Tax	Determined in equm.
b	0.40 $\cdot w$,	Social security benefits	Shimmer and Krause and Lubik (2005)
ψ	0	Share	
Δ	0	Credit spread	
$\bar{a}$	0	Borrowing constraint	

6 Baseline Results

This section summarizes the baseline equilibrium and the key mechanisms that discipline the quantitative analysis. In the baseline calibration, average entrepreneurship is 0.145 and average firm size is 21.877 workers, with strong heterogeneity across education groups. Entrepreneurship rates are 0.143 (low), 0.137 (middle), and 0.161 (high), while firm sizes are 9.682, 18.007, and 33.417 respectively. The cross-group gradient indicates that higher human capital is associated with both stronger entrepreneurial selection and larger operating scale.

The model also generates a positive but small necessity margin in the baseline (“Unemp-Ent” = 0.00204), consistent with a setting where labor market risk is present but not extreme. This margin is central for policy: it captures the share of entry that is induced by weak outside options rather than strong managerial comparative advantage.

The baseline serves as the reference point for all counterfactuals below. Comparisons to no-unemployment and alternative unemployment-insurance regimes show that policy affects not only total entrepreneurship, but also composition between necessity and opportunity entry. A key implication is that higher entrepreneurship rates are not mechanically welfare-improving; composition and firm-size responses are needed to interpret whether reallocation moves toward or away from efficiency.

Two baseline moments are especially useful for interpretation of later counterfactuals. First, the necessity-entry margin is 0.00204, which is about 1.41% of total entrepreneurship in the

baseline (0.145). Second, average firm size for high education is roughly 3.45 times the low-education value (33.417 versus 9.682), indicating that composition shifts across education types can generate large aggregate productivity effects even when total entry moves modestly.

7 Quantitative Analysis

This section reports policy counterfactuals relative to the baseline in Section 6. Current status is:

- Completed and interpreted: No-unemployment, UI variation (constant-tax), and selected consumption-floor specifications.
- Implemented but awaiting table completion: consumption-floor runs for $UI=0.0$ and low matching efficiency.
- Planned for next draft: credit-friction wedge and job-market deterioration experiments in a unified comparison table.

7.1 Experiment Map (Slides and Paper)

To keep the paper aligned with the current slides ('Slides_paper_v4.tex'), we use the following mapping:

- Counterfactual 1 in slides $\leftrightarrow$ **No Unemployment** (Table 2).
- Counterfactual 2 in slides $\leftrightarrow$ **Unemployment Insurance** (Table 3).
- Counterfactual 3 in slides $\leftrightarrow$ **Consumption Floor at $UI=0.4$** (Table 4).
- Planned next-step slide items $\leftrightarrow$ **Credit friction wedge** and **Job market experiment** subsections below.

7.2 No Unemployment

This counterfactual sets unemployment risk to zero, shutting down the necessity entry channel by construction. Relative to baseline, entrepreneurship rises from 0.145 to 0.197 and average firm size increases from 21.877 to 24.126. The increase in both entry and scale is consistent with stronger entrepreneurial selection when workers are not forced into low-outside-option states.

7.3 Unemployment Insurance

Constant-tax comparison We vary unemployment insurance generosity b while holding tax parameters fixed. In this specification, budget imbalances are treated as part of the counterfactual environment (that is, not forced to be revenue-neutral in each state).

Table 2: Counterfactual 1 (Slides): No unemployment risk

	Baseline (UI= 0.4)	No unemployment
entrepreneur-avg	0.145	0.197
entrepreneur-low	0.143	0.207
entrepreneur-mid	0.137	0.184
entrepreneur-high	0.161	0.216
firm size-avg	21.877	24.126
firm size-low	9.682	9.884
firm size-mid	18.007	19.814
firm size-high	33.417	37.214
Unemp-Ent	0.00204	0

Revenue-neutral comparison A revenue-neutral UI reform requires joint adjustment of b and tax rates. This specification is listed as pending and should be reported separately from the constant-tax results.

Table 3: Counterfactual 2 (Slides): UI level changes, constant-tax specification

	Baseline, UI=0.4	UI=0.3	UI=0.0	UI=0.6
entrepreneur-avg	0.145	0.145	0.193	0.144
entrepreneur-low	0.143	0.141	0.197	0.135
entrepreneur-mid	0.137	0.136	0.187	0.141
entrepreneur-high	0.161	0.164	0.181	0.154
firm size-avg	21.877	21.550	14.68	21.177
firm size-low	9.682	9.765	6.13	9.814
firm size-mid	18.007	17.931	11.33	17.023
firm size-high	33.417	32.477	23.98	32.958
Unemp-Ent	0.00204	0.00197	0.005	0.00084

The current constant-tax table shows that lower UI (UI= 0.0) raises measured entrepreneurship but reduces average firm size sharply, consistent with additional necessity entry. Higher UI (UI= 0.6) lowers necessity entry (“Unemp-Ent”) and has smaller effects on aggregate entrepreneurship.

7.4 Consumption floor

Policy design We evaluate two consumption-floor implementations around a minimum consumption threshold c : (i) a fiscal transfer that tops up low disposable income, and (ii) a reduced-form non-pecuniary penalty for falling below the threshold.

Interpretation For the completed specifications (UI= 0.4 and UI= 0.6), the effects on both entrepreneurship rates and firm size are quantitatively modest. This suggests that, at current calibrations, the consumption floor mainly smooths downside risk without materially changing occupational sorting.

Pending: fill this table from the UI= 0.0 baseline and corresponding consumption-floor runs.

Table 4: Counterfactual 3 (Slides): Consumption floor under baseline UI= 0.4

	Baseline, UI=0.4	Lump-sum transfer	Penalty
entrepreneur-avg	0.145	0.144	0.144
entrepreneur-low	0.143	0.143	0.142
entrepreneur-mid	0.137	0.136	0.137
entrepreneur-high	0.161	0.160	0.159
firm size-avg	21.877	21.867	21.951
firm size-low	9.682	9.655	9.776
firm size-mid	18.007	17.988	17.923
firm size-high	33.417	33.431	33.781
Unemp-Ent	0.00204	0.00189	0.00177

Table 5: Consumption floor under low UI benchmark (UI= 0.0) [pending fill]

	Baseline, UI=0.0	Lump-sum transfer	Penalty
entrepreneur-avg	0.145		
entrepreneur-low	0.143		
entrepreneur-mid	0.137		
entrepreneur-high	0.161		
firm size-avg	21.877		
firm size-low	9.682		
firm size-mid	18.007		
firm size-high	33.417		
Unemp-Ent	0.00204		

Table 6: Consumption floor under UI= 0.0 with lower matching efficiency [pending fill]

	Baseline, UI=0.0	Lump-sum transfer	Penalty
entrepreneur-avg			
entrepreneur-low			
entrepreneur-mid			
entrepreneur-high			
firm size-avg			
firm size-low			
firm size-mid			
firm size-high			
Unemp-Ent			

Pending: fill this table from the low-matching-efficiency counterfactuals.

Non-pecuniary penalty: Households face a utility penalty when consumption falls below $\underline{c}$. This reduced-form case is intended to capture environments where formal social protection is limited.

Table 7: Consumption floor under high UI benchmark (UI= 0.6)

	Baseline, UI=0.6	Lump-sum transfer	Penalty
entrepreneur-avg	0.144	0.146	0.145
entrepreneur-low	0.135	0.136	0.140
entrepreneur-mid	0.141	0.142	0.141
entrepreneur-high	0.154	0.156	0.155
firm size-avg	21.177	21.155	21.115
firm size-low	9.814	9.817	9.539
firm size-mid	17.023	17.023	17.056
firm size-high	32.958	32.885	32.793
Unemp-Ent	0.00084	0.0011	0.00117

7.5 Consumption floor with no unemployment or taxes

This counterfactual removes unemployment risk and taxes and then layers the consumption-floor variants on top. Results are nearly unchanged across columns, which is consistent with negligible binding of the floor in a high-insurance, no-unemployment environment.

Table 8: Consumption floor with no unemployment risk and no taxes

	no unemployment	Lump-sum transfer	Penalty
entrepreneur-avg	0.197	0.197	0.197
entrepreneur-low	0.207	0.207	0.206
entrepreneur-mid	0.184	0.184	0.184
entrepreneur-high	0.216	0.216	0.216
firm size-avg	24.126	24.127	24.131
firm size-low	9.884	9.884	9.903
firm size-mid	19.814	19.815	19.817
firm size-high	37.214	37.216	37.219
Unemp-Ent			

7.6 Credit friction wedge

This section is reserved for the credit-spread experiment that raises borrowing costs for entrepreneurs. The next draft should report: (i) entry rates by education and asset quantile, (ii) average firm size, and (iii) the necessity share relative to baseline.

7.7 Job market experiment

This section is reserved for labor market deterioration counterfactuals. The target comparison is a decline in matching efficiency and a decline in entrepreneurial productivity, with a common output table reporting levels and changes in entrepreneurship composition.

8 Conclusion

This paper develops a high-level quantitative framework for studying necessity entrepreneurship as an equilibrium outcome of occupational choice under labor market frictions. By defining necessity and opportunity entrepreneurship through counterfactual choices, the model separates compositional shifts in entry from changes in aggregate entrepreneurship rates.

The main message is that unemployment risk and social insurance jointly shape who becomes an entrepreneur. Reducing insurance can raise entrepreneurship on paper, but part of this increase reflects necessity entry associated with lower average firm scale. Conversely, stronger insurance can reduce necessity entry and improve sorting, even when effects on total entrepreneurship are modest. These results support evaluating entrepreneurship policy through composition, not levels alone.

Several extensions remain important for the next draft. First, the credit friction wedge and job-market experiments should be fully completed and integrated into a unified welfare comparison. Second, calibration choices that are still open should be aligned with the code implementation and reported transparently. Third, the literature discussion can be broadened further once the project-specific verified PDF library is restored.

Appendix

Computation

Given the values for parameters, the discretized spaces Ω_a for a , Ω_z for z and Ω_x for x . Each agent draw a human capital level h (interpreted as educational attainment) at the beginning of the economy, which will be fixed henceforth. h will determine their labor productivity $z \in \Omega_z$, the separation rate ρ , and managerial ability $x \in \Omega_x$. Let $i_w = 0$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 2$ being unemployed. The numerical algorithm works as follows.

1. Set a tolerance $\epsilon > 0$.
2. Guess $\Phi^0 = (r^0, w^0, \rho^0, \tau_y^0)$. Solve for optimal household behavior:

$$f : (\varphi; \Phi) \rightarrow (c, a', \mathcal{I}_e, n, k),$$

where $\varphi = \{h, z, x, a, i_w\}$. We will use the method of value function iteration as follows.

- (a) Guess value function $\mathbf{V}^0(\varphi; \Phi^0)$ and policy functions $f^0(\varphi; \Phi^0)$.
- (b) Calculate the cutoff values of x_u^* (for previously unemployed worker, or entrepreneur) and x_n^* (for previously employed worker) via the following conditions.

$$\tilde{w}_u = \pi(x_u^*) = \max_n \left\{ (x_u^*)^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1 + \psi\tau)n - rk - \frac{\kappa zn}{q(\theta)} \right\} \quad (11)$$

$$\tilde{w}_n = \pi(x_n^*) = \max_n \left\{ (x_n^*)^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1 + \psi\tau)n - rk - \frac{\kappa zn}{q(\theta)} \right\} \quad (12)$$

where $\tilde{w}_n$ and $\tilde{w}_u$ denote the expected compensation for the employed worker and unemployed worker respectively. More specifically,

$$\tilde{w}_n = (1 - \rho) \{zw(1 - (1 - \psi)\tau)\} + \rho b \quad (13)$$

$$\tilde{w}_u = p(\theta) \{zw(1 - (1 - \psi)\tau)\} p(\theta) b. \quad (14)$$

Noting that the separation rate ρ is pre-determined by the individual's initial human capital level h .

(c) Update value and policy functions:

$$\mathbf{V}^1(h, z, x, a, i_w) = \max_{\{a', c, \mathcal{I}_e\}} \left\{ \begin{array}{l} \iota_{i_w=0} \cdot \left[(1 - \mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] + \\ \iota_{i_w=1} \cdot \left[(1 - \mathcal{I}_e) \cdot \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] + \\ \iota_{i_w=2} \cdot \left[(1 - \mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] \end{array} \right\} \quad (15)$$

$$f^1(\varphi; \Phi^0) = \arg \max \mathbf{V}^1(\varphi; \Phi^0)$$

where $\mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w))$ denotes the expected payoff for the individual who is previously unemployed worker, or entrepreneur, and decides to look for paid job, while $\mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w))$ represents the expected payoff for the previously employed worker who decide to stay in workplace.

$$\mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w)) = p(\theta) (V_n + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 1)) \quad (16)$$

$$+ (1 - p(\theta)) (V_u + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 0))$$

$$\mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w)) = (1 - \rho) (V_n + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 1)) \quad (17)$$

$$+ \rho (V_u + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 0))$$

To simplify the analysis, we assume that the individual who was previously employed and newly separated from the firm is not allowed to search for job at the current period. The measure for the individuals who are looking for jobs, u_t , the total vacancies v_t , and job matching rate $\theta_{n,t}$ are given as follows.

$$u_t = \left(\int_x \iota_{i_w=0} + \int_x \iota_{i_w=2} \right) \cdot \int_0^{x_u^*} \quad (18)$$

$$v_t = \int_x \iota_{i_w=1} \cdot \rho \quad (19)$$

$$\theta_{n,t} = \bar{m} \left(\frac{v_t}{u_t} \right)^{1-\mu} \quad (20)$$

Note that we can write the above value functions explicitly as a system of sub-problems: $P_{x < x_u^*}^{i_w=0}$, $P_{x \geq x_u^*}^{i_w=0}$, $P_{x < x_n^*}^{i_w=1}$, $P_{x \geq x_n^*}^{i_w=1}$, $P_{x < x_u^*}^{i_w=2}$, $P_{x \geq x_u^*}^{i_w=2}$, which correspond to the dynamic programming problem for i). previously self-employed with $x < x_u^*$; ii). previously self-employed with $x \geq x_u^*$; iii). previously employed with $x < x_n^*$; iv). previously employed with $x \geq x_n^*$; v). previously unemployed with $x < x_u^*$; vi). previously unemployed with $x \geq x_u^*$. To economize the space, we only provide the details for

$P_{x < x_u^*}^{i_w=0}$, and $P_{x \geq x_u^*}^{i_w=0}$ as follows.
 $\left(P_{x < x_u^*}^{i_w=0}\right)$:

$$\mathbf{V}^1(h, z, x < x_u^*, a, i_w = 0) = \max_{a'_n, a'_u} \left\{ \begin{array}{l} p(\theta) (u(c_n) + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a'_n, i'_w = 1)) \\ + (1 - p(\theta)) (u(c_u) + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a'_u, i'_w = 0)) \end{array} \right\} \quad (21)$$

s.t.

$$c_n + a'_n \leq (1 - \tilde{r} - \delta)a + zw(1 - (1 - \psi)\tau), \text{ if matched with a firm} \quad (22)$$

$$c_u + a'_u \leq (1 - \tilde{r} - \delta)a + b, \text{ if unemployed} \quad (23)$$

$\left(P_{x \geq x_u^*}^{i_w=0}\right)$:

$$\mathbf{V}^1(h, z, x \geq x_u^*, a, i_w = 0) = \max_{a'_e} \{u(c_e) + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a'_e, i'_w = w)\} \quad (24)$$

s.t.

$$c_e + a'_e \leq (1 - \tilde{r} - \delta)a + \nu(k, x; \tilde{r}, w, \tau) \quad (25)$$

where

$$\nu(k, x; \tilde{r}, w, \tau) = \max_n \left\{ x^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1 + \psi\tau)n - rk - \frac{\kappa \rho n}{q(\theta)} \right\}. \quad (26)$$

(d) Stop if $\max \{|\mathbf{V}^1 - \mathbf{V}^0|, |f^1 - f^0|\} \leq \epsilon$. Set $\mathbf{V}^* = \mathbf{V}^1$, and $f^* = f^1$. Otherwise, set $\mathbf{V}^0 = \mathbf{V}^1$, $f^0 = f^1$ and repeat from step (c).

3. Generate a large number of individuals, $N = 100000$. For each agent j assign a vector of initial condition $(h_t^j, z_0^j, x_0^j, a_0^j, i_{w,0}^j)$, where $x_0^j \sim \Omega_x$, $z_0^j \in \Omega_z$, $i_{w,0}^j \in \Omega_w$.
4. Simulate the economy for T periods, where T is sufficiently large.
5. Calculate the following statistics from the simulated path

$$\left\{ h_t^j, z_t^j, x_t^j, a_t^j, i_{w,t}^j, \mathcal{I}_{e,t}^j, n_t^j, k_t^j \right\}_{t=0}^T.$$

$$LS^0 = \frac{\sum_{j=1}^N (\iota_{i_w=1} \cdot z^j)}{N} \quad (27)$$

$$KS^0 = \frac{\sum_{j=1}^N (\iota_{i_w=0} \cdot a^j + \iota_{i_w=1} \cdot a^j + \iota_{i_w=2} \cdot a^j - \iota_{i_w=0} \cdot k^j)}{\sum_{j=1}^N a^j} \quad (28)$$

$$u_t = \left(\int_h \iota_{i_w=0} + \int_h \iota_{i_w=2} \right) \cdot \int_0^{h_u^*} \quad (29)$$

$$v_t = \int_h \iota_{i_w=1} \cdot \rho \quad (30)$$

$$p(\theta)_t = \bar{m} \left(\frac{v_t}{u_t} \right)^{1-\mu} \quad (31)$$

$$G = \int \tau w z \quad (32)$$

and τ_y^1 that balances the government's budget.

6. Stop and set $(r^*, w^*, \theta^*) = (r^0, w^0, \theta^0)$, if $\max \{LS^0, KS^0, |\theta_n^1 - \theta_n^0|\} \leq \epsilon$. Otherwise, update aggregate variables (restart from step 2):

$$\begin{aligned} r^0 &= \chi r^0 + (1 - \chi) r^1 (KS^0) \\ w^0 &= \chi w^0 + (1 - \chi) w^1 (LS^0) \\ p(\theta)^0 &= \chi p(\theta)^0 + (1 - \chi) p(\theta)^1 \\ \tau_y^0 &= \chi \tau_y^0 + (1 - \chi) \tau_y^1 \end{aligned}$$

where $\chi \in (0, 1)$ is the step for updating aggregate variables.

Appendix Search and Matching

I am not sure how much of this is needed for coding purposes but this is the standard search and matching model. The stock of employed workers n_t can be expressed as the current pool of workers who did not separate from firms, plus new hires

$$n_t = (1 - \rho)n_{t-1} + m_{t-1}$$

The matching function is Cobb-Douglass where

$$m_t = \bar{m} v_t^{1-\mu} u_t^\mu$$

$\bar{m}$ is a match efficiency parameter, v is vacancies, u is unemployment, and match elasticity is $0 < \mu < 1$. The hiring rate can be defined for firms as total number of new hires to the number of vacancies

$$q(\theta_t) = \frac{m_t}{v_t} = \bar{m}\theta_t^{-\mu}$$

where $\theta_t = \frac{v_t}{u_t}$ is aggregate labour market tightness. Similarly we can define this from the perspective of the worker in the job-finding rate.

$$p(\theta_t) = \frac{m_t}{u_t} = \bar{m}\theta_t^{1-\mu}$$

The Job Creation Condition determines the amount of new vacancies an employer will post. In our model, most workers are on rolling contracts, so firms only need to post new vacancies after some workers separate from these rolling contracts. Usually in search and matching models the wage is determined ex-post of matching by Nash bargaining between the employer and employee. However, in our span-of-control model the wage has already been determined ex ante, as the wage is an endogenous variable that determines occupational choice. The marginal benefit that the new employee would bring to the firm is expressed as A (which can be derived from the firms production function, $\gamma X k^\alpha (zn)^{\gamma-1}$). The *JCC* can be then expressed as the costs of replacing out workers on the *LHS* and the expected benefit on the *RHS*

$$\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t \left[A - w_{t+1} + (1 - \rho) \frac{\kappa}{q(\theta_{t+1})} \right]$$

where κ is the cost of posting a vacancy.

Matching Steady States

To find the steady state of the employment equation $n_t = (1 - \rho)n_{t-1} + m_{t-1}$, we use the fact that $u = (1 - n)$ and $m_t = \bar{m}v_t^{1-\mu}u_t^\mu$. Substituting and getting rid of time subscript gives $(1 - u) = (1 - \rho)(1 - u) + \bar{m}v^{1-\mu}u^\mu$ which simplifies to $\bar{m}v^{1-\mu}u^\mu = \rho(1 - u)$. Making v the subject of the formula gives us the Beveridge curve.

$$v = \left(\frac{\rho(1 - u)}{\bar{m}u} \right)^{1-\mu} u$$

The steady state for the JCC can be found by using $\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t \left[A - w_{t+1} + (1 - \rho) \frac{\kappa}{q(\theta_{t+1})} \right]$ and $q(\theta_t) = \bar{m}\theta_t^{-\mu} = \bar{m} \left(\frac{v_t}{u_t} \right)^{-\mu}$. Substituting and getting rid of time subscripts yields

$$\frac{\kappa}{\bar{m}} \left(\frac{v}{u} \right)^\mu = A - w_t + \beta \left[A - w + (1 - \rho) \frac{\kappa}{\bar{m}} \left(\frac{v}{u} \right)^\mu \right]$$

Simplifying yields the steady state of the JCC

$$1 - \beta(1 - \rho) \frac{\kappa}{\bar{m}} \left(\frac{v}{u} \right)^\mu = (1 + \beta) [A - w]$$

References

- Buera, F. J. (2009). A dynamic model of entrepreneurship with borrowing constraints: theory and evidence. *Annals of Finance*, 5(3–4):443–464.
- Fairlie, R. and Fossen, F. (2019). *Defining Opportunity versus Necessity Entrepreneurship: Two Components of Business Creation*.
- Hurst, E. and Pugsley, B. W. (2011). What do small businesses do? *Brookings Papers on Economic Activity*, 2011(2):73–118.
- Petrongolo, B. and Pissarides, C. A. (2001). Looking into the black box: A survey of the matching function. *Journal of Economic Literature*, 39(2):390–431.
- Sedlacek, P. and Sterk, V. (2017). The growth potential of startups over the business cycle. *American Economic Review*, 107(10):3182–3210.